RAHUL

Glasgow, United Kingdom | rahul@email.com | +44 7763338784
linkedin.com/in/rahul |
github.com/rahul

PERSONAL DETAILS

Location: Glasgow, United Kingdom | **Work Eligibility:** UK Student Visa

Nationality: Indian | **Availability:** Open to AI/ML internships and junior roles

PROFESSIONAL SUMMARY

MSc Robotics & AI candidate at the University of Glasgow with hands-on experience building production-grade agentic AI systems using LangChain, CrewAI, and Model Context Protocol (MCP). Proven track record delivering LLM-powered chatbots, RAG pipelines, and deep learning models across real-world projects including a 60–75% reduction in client response time at Epic Minds. Seeking ML/AI Engineer internship or junior roles in the UK.

EDUCATION

Master of Science in Robotics and Artificial Intelligence *Sept 2025 – Present*
University of Glasgow, Glasgow, UK

Bachelor of Engineering in Computer Science and Engineering *Aug 2021 – June 2025*
Global Academy of Technology, Bangalore, India

TECHNICAL SKILLS

Languages: Python, SQL, C++

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers

Agentic AI & LLMs: LangChain, CrewAI, Agno (formerly Phidata), Model Context Protocol (MCP), RAG Architectures, Prompt Engineering

Web & API: FastAPI, Streamlit, REST APIs, HTML/CSS

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Cloud & DevOps: AWS Bedrock, Git, GitHub, Docker

INTERNSHIPS

Python Developer Intern *April 2025 – Sept 2025*
Epic Minds, Bangalore, India

- Built AI-powered enterprise applications using Python, FastAPI, LangChain, and Model Context Protocol (MCP), enabling autonomous tool integration and workflow automation.
- Developed a context-aware conversational chatbot for a consultancy firm, reducing average client response time by **60–75%** through retrieval-augmented and personalized interactions.
- Designed RESTful APIs and collaborated with front-end engineers to integrate AI services into production applications, improving scalability and end-to-end user experience.

PROJECTS

GeneWeaver v3 – Multi-Agent Protein Function Annotation System (MSc Dissertation) *May 2026 – Present*

- Designing and developing a proof-of-concept multi-agent bioinformatics system that automatically annotates protein function from raw DNA sequences using cooperative AI agents orchestrated with LangGraph.
- Building an end-to-end genome annotation pipeline incorporating gene prediction, protein translation, Protein Language Models (ESM-2), biological database retrieval (BLAST, UniProt, InterPro, PDB), and scientific literature analysis (PubMed).
- Implementing a consensus reasoning agent that integrates evidence from multiple independent agents to resolve conflicting predictions and generate confidence-scored Gene Ontology (GO) functional annotations.
- Evaluating the system using the CAFA benchmarking methodology with SwissProt time-split datasets, Fmax and Smin metrics, ablation studies, and comparisons against BLAST and Prokka on *Pseudomonas aeruginosa* PA14 and *E. coli* K-12 benchmark genomes.

Stock Price Prediction – GRU with Temporal Attention *Feb 2026 – Apr 2026*

- Challenged to predict next-day closing prices for 442 stocks simultaneously across 3,021 trading days, requiring a scalable sequence model capable of capturing cross-stock temporal dependencies.
- Engineered a GRU-based architecture with a learnable temporal attention mechanism, per-stock Z-score normalization, and a sliding-window sequence pipeline, optimizing seven hyperparameters across 50 Optuna TPE trials with median pruning.
- Integrated Captum Integrated Gradients for feature attribution and interpretability, visualizing which historical time steps most influenced predictions to validate the attention mechanism's learned weighting.
- Achieved a best validation MSE of **1.8893** after automated hyperparameter search across 50 Optuna trials, with the attention-enabled architecture selected as the best-performing model.

Fourier Regression & Convolutional Autoencoder – Dimensionality Reduction Study *Jan 2026 – Mar 2026*

- Assigned to solve two PyTorch-based problems: approximating a periodic function via learned Fourier regression, and building a convolutional autoencoder for MNIST dimensionality reduction and downstream classification.
- Implemented a trainable Fourier expansion model that learned optimal a_k and b_k coefficients through gradient-based regression rather than analytical FFT, and constructed a Conv2d/ConvTranspose2d autoencoder with multiple latent-space dimensions.
- Compared reconstruction quality and classification accuracy of convolutional autoencoder encodings against PCA baselines using logistic regression on the latent representations.
- Achieved **92.69%** classification accuracy using autoencoder encodings versus **90.06%** with PCA at matched latent dimensions, evidencing the representational advantage of learned nonlinear embeddings.

Brain Tumor MRI Classification – Deep Learning for Medical Imaging *Oct 2025 – Nov 2025*

- Tasked with building a CNN-based classifier to categorize brain MRI scans into four tumor types across a dataset of 7,023 images (5,712 train / 1,311 validation) for a Deep Learning coursework at the University of Glasgow.
- Designed and trained a custom convolutional neural network in PyTorch, applying data augmentation, batch normalization, and transfer learning to address class imbalance and limited data in a mini-dataset variant.
- Evaluated model robustness by comparing full-dataset versus reduced-dataset performance, analyzing overfitting behaviour and generalization through confusion matrices and per-class accuracy metrics.
- Achieved **98.40%** validation accuracy on the full dataset and benchmarked custom CNN and ResNet50 transfer-learning models on a reduced dataset, demonstrating strong performance across varying data availability.

CERTIFICATIONS

AWS Academy Graduate – AWS Academy AI Foundations *June 2026*
Amazon Web Services (AWS) *Credential*

- Completed hands-on training in generative AI, prompt engineering, foundation models, and AWS AI services including Amazon Bedrock and SageMaker.

ACHIEVEMENTS

- Won the **FemEng Wind Farm Challenge** at the University of Glasgow (April 2026), leading GIS-based constraint mapping, OpenWind energy modelling, and financial analysis for a UK wind farm site that outperformed competing locations on a 25-year CAPEX and payback assessment.
- Secured **2nd Place (Runner-Up)** at the James Watt School of Engineering “Engineer Your Future: Challenge Day”, designing a modular wind turbine blade system with structural health monitoring and circular lifecycle strategies for ScottishPower Renewables, with special appreciation from ESBF.
- Secured **1st Place** in the “ML Matrix” Hackathon organized by the IEEE Society, competing against 50+ teams in machine learning challenges.
- Secured **3rd Place** in a corporate-level hackathon hosted by Rocketium, competing against industry professionals and delivering a production-grade AI ad-generation prototype.